CHEM 1043 General Chemistry II

Topic 4

Solubility and Complex-Ion Equilibria



(Class Practice Solutions)

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Class Practice 1

Solubility Product Constant Expression for Slightly Soluble Solutes

Write the solubility product constant expression for the solubility equilibrium of

(a) Calcium fluoride, CaF_2 (one of the substances used when a fluoride treatment is applied to teeth).

(b) Copper arsenate, $Cu_3(AsO_4)_2$ (used as an insecticide and fungicide).

Solution:

(a) $CaF_2(s) \rightleftharpoons Ca^{2+}(aq) + 2\,F^-(aq) \qquad K_{sp} = [Ca^{2+}][F^-]^2$

(b) $Cu_3(AsO_4)_2(s) \rightleftharpoons 3\,Cu^{2+}(aq) + 2\,AsO_4^{3-}(aq) \qquad K_{sp} = [Cu^{2+}]^3[AsO_4^{3-}]^2$

Class Practice 2

Calculating K_{sp} of a Slightly Soluble Solute from Its Solubility

A handbook lists the aqueous solubility of AgOCN as 7 mg/100 mL at 20 °C. What is the K_{sp} of AgOCN at 20 °C?

Solution:

First calculate the solubility of silver cyanate, *s*, as a molarity.
Then use the solubility equation to
(1) relate the concentrations of the ions and
(2) write the K_{sp} expression.

$$s = \frac{7\ \text{mg AgOCN}}{100\ \text{mL}} \times \frac{1000\ \text{mL}}{1\ \text{L}} \times \frac{1\ \text{g}}{1000\ \text{mg}} \times \frac{1\ \text{mol AgOCN}}{149.9\ \text{g AgOCN}} = 5 \times 10^{-4}\ \text{M}$$

Equation : $\text{AgOCN}(s) \rightleftharpoons \text{Ag}^+(aq) + \text{OCN}^-(aq)$

Solubility Product : s s

$$K_{sp} = [\text{Ag}^+][\text{OCN}^-] = (s) \times (s) = s^2 = (5 \times 10^{-4})^2 = 3 \times 10^{-7}$$

Class Practice 3

Calculating the Solubility of a Slightly Soluble Solute from Its K_{sp} Value

The K_{sp} of $Cu_3(AsO_4)_2$ at 25 °C is 7.6×10^{-36}. What is the molar solubility of $Cu_3(AsO_4)_2$ in H_2O at 25 °C?

Solution:

Use the solubility equilibrium to write the K_{sp} expression, which we then solve to obtain the molar solubility, s, of $Cu_3(AsO_4)_2$.

$$Cu_3(AsO_4)_2.(s) \rightleftharpoons 3\ Cu^{2+}(aq) + 2\ AsO_4^{3-}(aq)$$

$$K_{sp} = \left[Cu^{2+}\right]^3 \left[AsO_4^{3-}\right]^2 = (3s)^3 (2s)^2 = 108s^5 = 7.6 \times 10^{-36}$$

$$\text{Solubility :} \qquad s = \sqrt[5]{\frac{7.6 \times 10^{-36}}{108}} = 3.7 \times 10^{-8}\ \text{M}$$

Class Practice 4

Calculating the Solubility of a Slightly Soluble Solute in the Presence of a Common Ion

What is the molar solubility of PbI_2 in 0.10 M $Pb(NO_3)_2$ (aq)?
(Given that $K_{sp} = 7.1 \times 10^{-9}$)
[*Hint:* To which ion concentration should the solubility be related?]

Solution:

The solubility equilibrium is the basis of the calculation.

	$PbI_2(s)$	$\rightleftharpoons$	$Pb^{2+}(aq)$	+	$2I^-(aq)$
Equation:					
Initial:	—		0.10 M		0 M
Changes:	—		$+s$ M		$+2s$ M
Equil:	—		$(0.10 + s)$ M		$2s$ M

$$K_{sp} = [Pb^{2+}][I^-]^2 = 7.1 \times 10^{-9} = (0.10 + s)(2s)^2 \approx 0.40\,s^2 \qquad s = \sqrt{\frac{7.1 \times 10^{-9}}{0.40}} = 1.3 \times 10^{-4}\ \text{M}$$

(assumption $0.10 \gg s$ is valid)

This value of s is the solubility of PbI_2 in 0.10 M $Pb(NO_3)_2$(aq).

Class Practice 5

Applying the Criteria for Precipitation of a Slightly Soluble Solute

Three drops of 0.20 M KI are added to 100.0 mL of a 0.010 M solution of $AgNO_3$. Will a precipitate of silver iodide form?

(Assume 1 drop = 0.05 mL, given that K_{sp} of AgI at 25 °C = 8.5×10^{-17}.)

Solution:

First determine $[I^-]$ as altered by dilution. Then calculate Q_{sp} and compare it with K_{sp}.

$$[I^-] = \frac{3\,\text{drops} \times \frac{0.05\,\text{mL}}{1\,\text{drop}} \times \frac{0.20\,\text{mmol KI}}{1\,\text{mL}} \times \frac{1\,\text{mmol I}^-}{1\,\text{mmol KI}}}{100.0\,\text{mL soln}} = 3 \times 10^{-4}\,\text{M}$$

$$Q_{sp} = [Ag^+][I^-] = (0.010\ )\left(3 \times 10^{-4}\right) = 3 \times 10^{-6}$$

$\because Q_{sp} > K_{sp}$, precipitation of AgI should occur.

Class Practice 6

Assessing the Completeness of a Precipitation Reaction

A typical Ca^{2+} concentration in seawater is 0.010 M. Will the precipitation of $Ca(OH)_2$ be complete from a seawater sample in which $[OH^-]$ is maintained at 0.040 M? Given K_{sp} of $Ca(OH)_2$ at 25 °C = 5.5×10^{-6}.

Solution:

First we find the maximum $[Ca^{2+}]$ that can coexist with $[OH^-] = 0.040$ M.

$$K_{sp} = 5.5\times10^{-6} = \left[Ca^{2+}\right]\left[OH^-\right]^2 = \left[Ca^{2+}\right](0.040)^2$$

$$\left[Ca^{2+}\right] = \frac{5.5\times10^{-6}}{(0.040)^2} = 3.4\times10^{-3}\,\text{M}$$

For precipitation to be considered complete, $[Ca^{2+}]$ should be less than 0.1% of its original value. From the result, 3.4×10^{-3} M is 34% of 0.010 M , which is **NOT** less than 0.1% .

Therefore, precipitation of $Ca(OH)_2$ is **NOT** complete under these conditions.

Class Practice 7

Separating Ions by Fractional Precipitation

$AgNO_3(aq)$ is slowly added to a solution with $[Cl^-]$ = 0.115 M and $[Br^-]$ = 0.264 M . What percent of the Br^- remains unprecipitated at the point at which AgCl(s) begins to precipitate?

Given that K_{sp} of AgCl = 1.8×10^{-10} and K_{sp} of AgBr = 5.0×10^{-13}.

Solution:

Let us first determine $[Ag^+]$ when AgCl(s) just begins to precipitate. At this point, Q_{sp} and K_{sp} are **equal**.

$$K_{sp} = 1.8\times10^{-10} = [Ag^+][Cl^-] = [Ag^+]\times 0.115\text{M} \qquad [Ag^+] = \frac{1.8\times10^{-10}}{0.115} = 1.6\times10^{-9}\,\text{M}$$

Next, determine the maximum $[Br^-]$ that can coexist with this $[Ag^+]$.

$$K_{sp} = 5.0\times10^{-13} = [Ag^+][Br^-] = 1.6\times10^{-9}\,\text{M}\times[Br^-];\ [Br^-] = \frac{5.0\times10^{-13}}{1.6\times10^{-9}} = 3.1\times10^{-4}\,\text{M}$$

The remaining bromide ion has precipitated as AgBr(s) with the addition of $AgNO_3$ (aq).

$$\text{Percent of } Br^- \text{ remaining} = \frac{[Br^-]_{final}}{[Br^-]_{initial}}\times100\% = \frac{3.1\times10^{-4}\,\text{M}}{0.264\,\text{M}}\times100\% = 0.12\%$$

As precipitation of AgBr is **NOT complete** when AgCl begins to precipitate, the fractional precipitation is NOT feasible

Class Practice 8

Determining Whether a Precipitate Will Form in a Solution in Which There Is Also an Ionization Equilibrium

Should $Mg(OH)_2(s)$ precipitate from a solution that is 0.010 M $MgCl_2(aq)$ and also 0.10 M $CH_3COO^-Na^+$?

$K_{sp}\,[Mg(OH)_2] = 1.8 \times 10^{-11}$; $K_a\,[CH_3COOH] = 1.8 \times 10^{-5}$

Solution:

Equation:	$C_2H_3O_2^-(aq)$	+ $H_2O(l)$	$\rightleftharpoons$	$HC_2H_3O_2(aq)$	+	$OH^-(aq)$
Initial:	0.10 M	—		0 M		≈ 0 M
Changes:	$-x$ M	—		$+x$ M		$+x$ M
Equil:	$(0.10 - x)$ M	—		x M		x M

$$K_b = \frac{1.0\times10^{-14}}{1.8\times10^{-5}} = 5.6\times10^{-10} = \frac{[HC_2H_3O_2][OH^-]}{[C_2H_3O_2^-]} = \frac{x\cdot x}{0.10-x} \approx \frac{x^2}{0.10}$$

$$x = [OH^-] = \sqrt{0.10\times5.6\times10^{-10}} = 7.5\times10^{-6}\,\text{M} \quad \text{(the assumption } x \ll 0.10 \text{ was valid)}$$

$$Q_{sp} = [Mg^{2+}][OH^-]^2 = (0.010\text{M})(7.5\times10^{-6}\text{M})^2 = 5.6\times10^{-13} < 1.8\times10^{-11} = K_{sp}[Mg(OH)_2]$$

As a result, precipitation does not occur.

Class Practice 9

Controlling an Ion Concentration, Either to Cause Precipitation or to Prevent It

What minimum [NH_4^+] must be present to prevent precipitation of $Mn(OH)_2(s)$ from a solution that is 0.0050 M $MnCl_2$ and 0.025 M NH_3?

K_{sp} [$Mn(OH)_2$] = 1.9×10^{-13}; pK_b [NH_3] = 4.74

Solution:

$$K_{sp} = 1.9\times10^{-13} = \left[Mn^{2+}\right]\left[OH^-\right]^2 = (0.0050\text{M})\left[OH^-\right]^2$$

$$\left[OH^-\right] = \sqrt{\frac{1.9\times10^{-13}}{0.0050}} = 6.2\times10^{-6}\ \text{M}$$

pK_b = - log[K_b], $K_b = 1.8 \times 10^{-5}$

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq) \qquad K_b = 1.8 \times 10^{-5}$$

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]} = \frac{[NH_4^+](\ 6.2\times10^{-6}\ )}{0.025} = 1.8 \times 10^{-5}$$

[NH_4^+] = 0.073 M

[NH_4^+] should be maintained at 0.073 M or higher

Class Practice 10

Determining Whether a Precipitate Will Form in a Solution Containing Complex Ions

Will AgCl(s) precipitate from 1.50 L of a solution that is 0.100 M $AgNO_3$ and 0.225 M NH_3 if 1.00 mL of 3.50 M NaCl is added? Given that K_f of $[Ag(NH_3)_2]^+ = 1.6 \times 10^7$.

Solution:

Equation: $Ag^+(aq) + 2\,NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+(aq)$

	$Ag^+(aq)$	$2\,NH_3(aq)$	$[Ag(NH_3)_2]^+(aq)$
In soln	0.100 M	0.225 M	0 M
Form complex	–0.100 M	–0.200 M	+0.100 M
Initial	0 M	0.025 M	0.100 M
Changes	+x M	+2x M	–x M
Equil	x M	(0.025 + x) M	(0.100 – x) M

$$K_f = 1.6\times10^7 = \frac{\left[[Ag(NH_3)_2]^+\right]}{[Ag^+][NH_3]^2} = \frac{0.100 - x}{x(0.025+2x)^2} \approx \frac{0.100}{x(0.025)^2}$$

$$x = \frac{0.100}{(0.025)^2 1.6\times10^7} = 1.0\times10^{-5}\,\text{M} = [Ag^+] = \text{concentration of free silver ion}$$

(x << 0.025 M, so the approximation was valid)

The $[Cl^-]$ is diluted: $[Cl^-]_{final} = [Cl^-]_{initial} \times \frac{1.00\,\text{mL}_{initial}}{1,500\,\text{mL}_{final}} = 3.50\,\text{M} \div 1500 = 0.00233\,\text{M}$

$$Q_{sp} = [Ag^+][Cl^-] = (1.0\times10^{-5}\,\text{M})(0.00233\,\text{M}) = 2.3\times10^{-8} > 1.8\times10^{-10} = K_{sp}$$

As $Q_{sp} > K_{sp}$, precipitation of AgCl(s) should occur.

Class Practice 11

Controlling a Concentration to Cause or Prevent Precipitation from a Solution of Complex Ions

What is the *minimum* concentration of NH_3 needed to prevent AgCl (s) from precipitating from 1.00 L of a solution containing 0.10 mol $AgNO_3$ and 0.010 mol NaCl?
(Given that K_f of $[Ag(NH_3)_2]^+ = 1.6 \times 10^7$ and K_{sp} of $AgCl = 1.8 \times 10^{-10}$.)

Solution:

Solve for the amount of silver ion that can be in solution without precipitation in a solution containing 1.0×10^{-2} M Cl^-. That is, $[Ag^+][Cl^-] \leq K_{sp}$.

$$[Ag^+](1.0 \times 10^{-2}) \leq K_{sp} = 1.8 \times 10^{-10} \qquad [Ag^+] \leq 1.8 \times 10^{-8}\,\text{M}$$

Thus, the maximum concentration of *uncomplexed* Ag^+ in solution is 1.8×10^{-8} M. This means that essentially all the Ag^+ (0.10 mol/L) must be tied up (complexed) in the complex ion $[Ag(NH_3)_2]^+$. We need to solve the expression at the right for $[NH_3]$.

$$K_f = \frac{[[Ag(NH_3)_2]^+]}{[Ag^+][NH_3]^2} = \frac{1.0 \times 10^{-1}}{1.8 \times 10^{-8}[NH_3]^2} = 1.6 \times 10^7$$

$$[NH_3]^2 = \frac{1.0 \times 10^{-1}}{1.8 \times 10^{-8} \times 1.6 \times 10^7} = 0.35 \qquad [NH_3] = 0.59\,\text{M}$$

The concentration just calculated is that of *free, uncomplexed* NH_3. Considering as well the 0.20 mol NH_3/L complexed in the 0.10 M $[Ag(NH_3)_2]^+$, we see that the total concentration of NH_3(aq) required is

$$[NH_3]_{tot} = 0.59\,\text{M} + 0.20\,\text{M} = 0.79\,\text{M}$$

Class Practice 12

Determining the Solubility of a Solute When Complex Ions Form

What is the molar solubility of AgCl in 0.100 M NH_3 (aq)?
(Given that K_f of $[Ag(NH_3)_2]^+ = 1.6 \times 10^7$ and K_{sp} of AgCl $= 1.8 \times 10^{-10}$.)

Solution

$$AgCl(s) + 2\,NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+(aq) + Cl^-(aq)$$

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq) \qquad K_{sp} = 1.8 \times 10^{-10}$$

$$Ag^+(aq) + 2\,NH_3(aq) \rightleftharpoons [Ag(NH_3)_2^+](aq) \qquad K_f = 1.6 \times 10^7$$

$$AgCl(s) + 2\,NH_3(aq) \rightleftharpoons [Ag(NH_3)_2^+](aq) + Cl^-(aq)$$

$$K = K_{sp} \times K_f = 1.8 \times 10^{-10} \times 1.6 \times 10^7 = 2.9 \times 10^{-3}$$

$$K = \frac{[[Ag(NH_3)_2]^+][Cl^-]}{[NH_3]^2} = \frac{s \cdot s}{(0.100 - 2s)^2} = \left(\frac{s}{0.100 - 2s}\right)^2 = 2.9 \times 10^{-3}$$

$$\frac{s}{0.100 - 2s} = \sqrt{2.9 \times 10^{-3}} = 5.4 \times 10^{-2}$$

$$s = 5.4 \times 10^{-3} - 0.11s$$
$$1.11s = 5.4 \times 10^{-3}$$
$$s = 4.9 \times 10^{-3}$$

The molar solubility of AgCl(s) in 0.100 M NH_3 (aq) is 4.9×10^{-3} M.

Class Practice 13

Suppose 0.100 mol of $AgNO_3$ is dissolved in 1.00 L of a 1.00 M NH_3 (aq). Calculate the concentrations of Ag^+ and $[Ag(NH_3)]^+$ ions present at equilibrium.

Given the following:

(step 1) $Ag^+(aq) + NH_3(aq) \rightleftharpoons Ag(NH_3)^+(aq)$ $\qquad \dfrac{[Ag(NH_3)^+]}{[Ag^+][NH_3]} = K_1 = 2.1 \times 10^3$

(step 2) $Ag(NH_3)^+(aq) + NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+(aq)$ $\qquad \dfrac{[Ag(NH_3)_2^+]}{[Ag(NH_3)^+][NH_3]} = K_2 = 8.2 \times 10^3$

Solution:

Suppose most of the Ag^+ is present as $Ag(NH_3)_2^+$. Then $[Ag(NH_3)_2^+]_0 = 0.100$ M and $[NH_3]_0 = 0.80$ M.

Let y mol L^{-1} of $Ag(NH_3)_2^+$ dissociates at equilibrium:

	$Ag(NH_3)_2^+(aq) \rightleftharpoons$	$Ag(NH_3)^+(aq)$ +	$NH_3(aq)$
Initial concentration (M)	0.100	0	0.80
Change in concentration (M)	$-y$	$+y$	$+y$
Equilibrium concentration (M)	$0.100 - y$	y	$0.80 + y$

Class Practice 13 (Continued)

then the first equilibrium expression becomes

$$\frac{y(0.80 + y)}{0.10 - y} = \frac{1}{K_2} = \frac{1}{8.2 \times 10^3}$$

$$y = 1.5 \times 10^{-5}\ \text{M} = [Ag(NH_3)^+]$$

We can then calculate the concentration of free Ag^+ ions as follows:

$$\frac{[Ag^+][NH_3]}{[Ag(NH_3)^+]} = \frac{1}{K_1}$$

$$\frac{[Ag^+](0.80)}{1.5 \times 10^{-5}} = \frac{1}{2.1 \times 10^3}$$

$$[Ag^+] = 9 \times 10^{-9}\ \text{M}$$

It is clear that the original assumption was correct, and most of the silver present is tied up in the $Ag(NH_3)_2^+$ complex.

Class Practice 14

Separating Metal Ions by Selective Precipitation of Metal Sulfides

Show that $Ag_2S(s)$ ($K_{spa} = 6 \times 10^{-30}$) should precipitate and that FeS(s) ($K_{spa} = 6 \times 10^{2}$) should not precipitate from a solution that is 0.010 M Ag^+ and 0.020 M Fe^{2+}, and maintained with $[H_3O^+] = 0.30$ M.

Solution:

First, determine the value of Q_{spa} in each case, with $[H_2S] = 0.10$ M and $[H_3O^+] = 0.30$ M.

$$FeS\,(s) + 2H_3O^+\,(aq) \rightleftharpoons Fe^{2+}\,(aq) + H_2S\,(aq) + 2H_2O\,(l)$$

$$\text{For FeS, } Q_{spa} = \frac{0.020 \times 0.10}{(0.30)^2} = 0.022 < 6 \times 10^2 = K_{spa}$$

Thus, precipitation of FeS should not occur.

$$Ag_sS\,(s) + 2H_3O^+\,(aq) \rightleftharpoons 2Ag^+\,(aq) + H_2S\,(aq) + 2H_2O\,(l)$$

$$\text{For } Ag_2S, Q_{spa} = \frac{(0.010)^2 \times 0.10}{(0.30)^2} = 1.1 \times 10^{-4}$$

$Q_{spa} > 6 \times 10^{-30} = K_{spa}$; thus, precipitation of Ag_2S should occur.

Class Practice 15

Both Cu^{2+} and Ag^{+} are present in the same aqueous solution. Explain which of the following reagents would work best in separating these ions, precipitating one and leaving the other in solution: $(NH_4)_2CO_3(aq)$, $HNO_3(aq)$, $H_2S(aq)$, $HCl(aq)$, $NH_3(aq)$, or $NaOH(aq)$.

Solution:

- $(NH_4)_2CO_3(aq)$, $H_2S(aq)$, and $NaOH\ (aq)$ do not work because both ions would precipitate.
- $HNO_3(aq)$ and $NH_3(aq)$ do not work because neither ion would precipitate.
- $HCl(aq)$ works because $CuCl_2$ is soluble in water but AgCl is not.

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